

Problema 2: SISTEMA DE MONITORAMENTO COM CONTAGEM DE EVENTOS

TEC498-TP01(2026.1)

Discentes:
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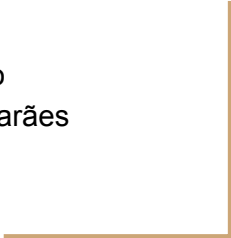
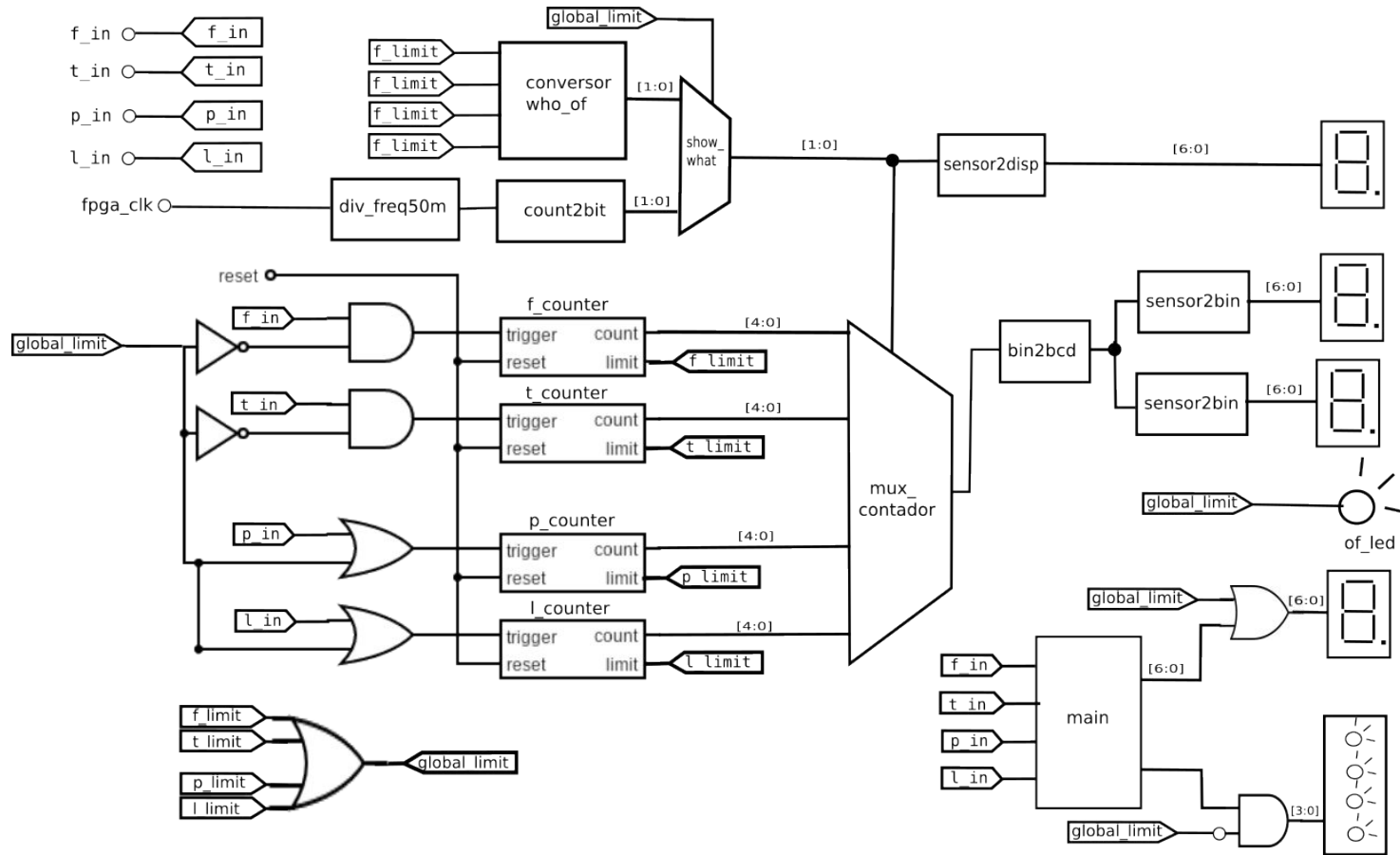
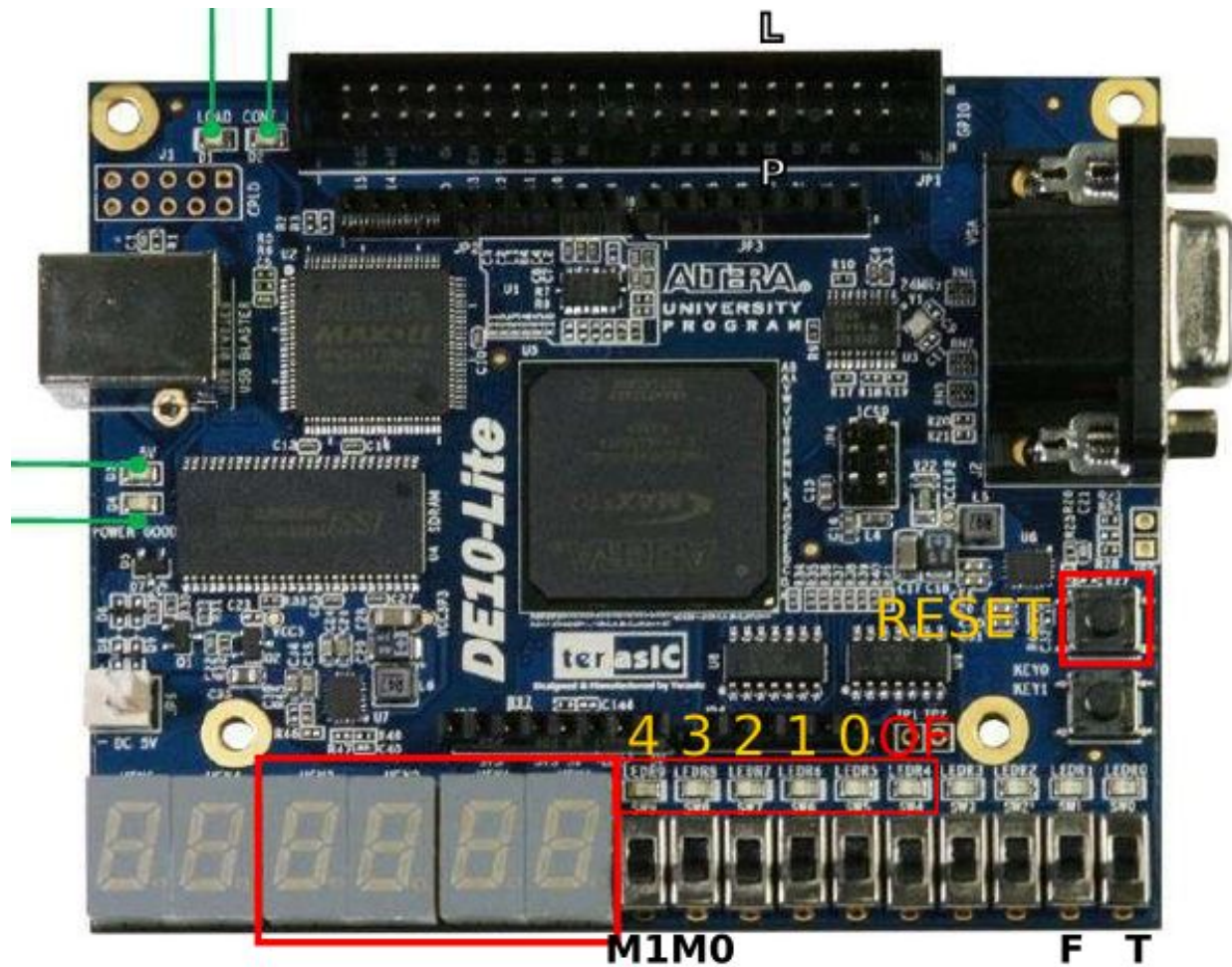


DIAGRAMA DE ALTO NÍVEL





SIGNIFICADO DOS LEDS

Modo Normal

LED 2 - Perigo

LED 1 - Invasão

LED 0 - Alerta

Modo Diagnóstico

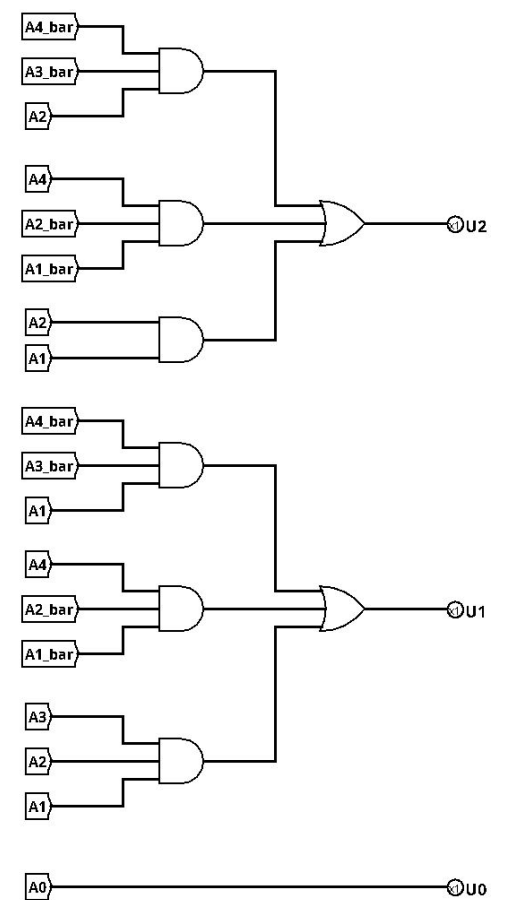
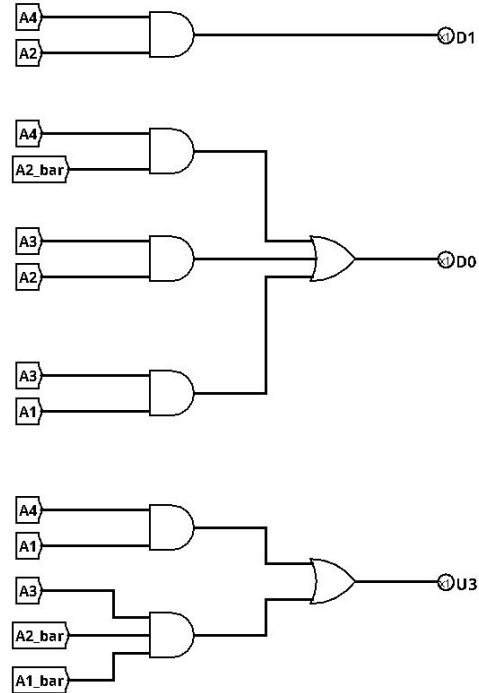
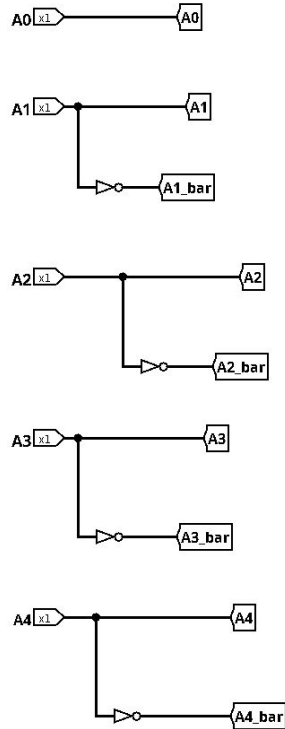
LED 3 - MSB, LED 0 - LSB

Modo Manutenção

Todos os LEDs acendem

Sistema travado

LED OF - of_led



Desenho Do Circuito BINÁRIO 5 BITS PARA BCD

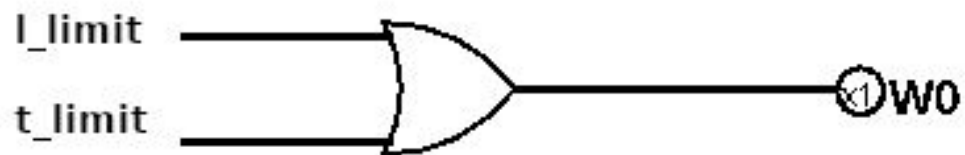
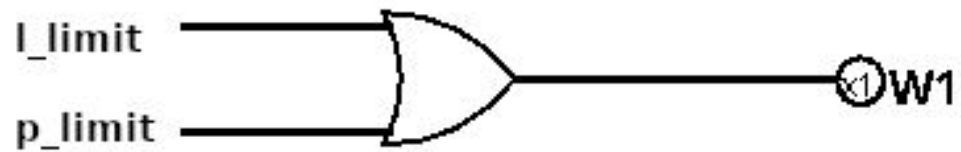
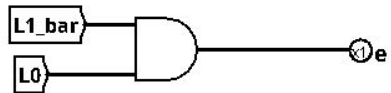
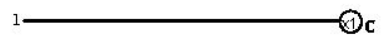
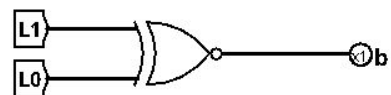
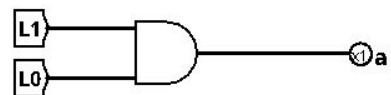
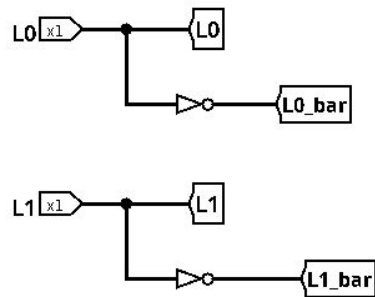


Diagrama do componente `conv_who_overflow`



Desenho LETRA SENSOR → DISPLAY

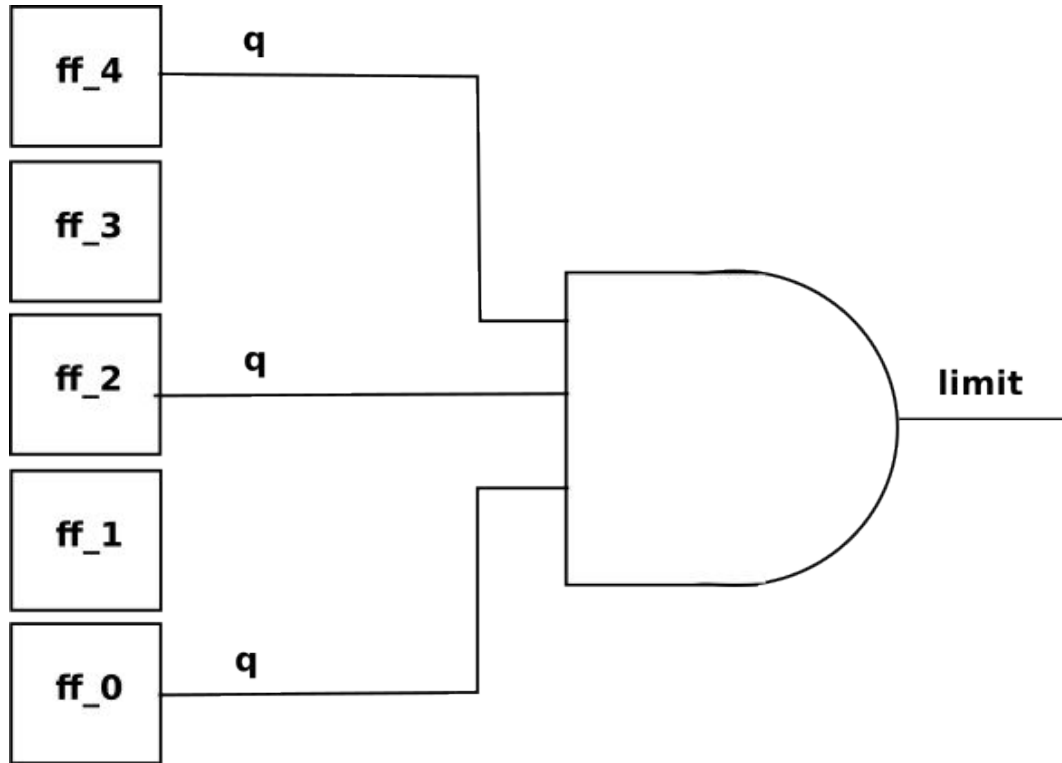


Diagrama simplificado do mecanismo de aviso de “estouro” do contador

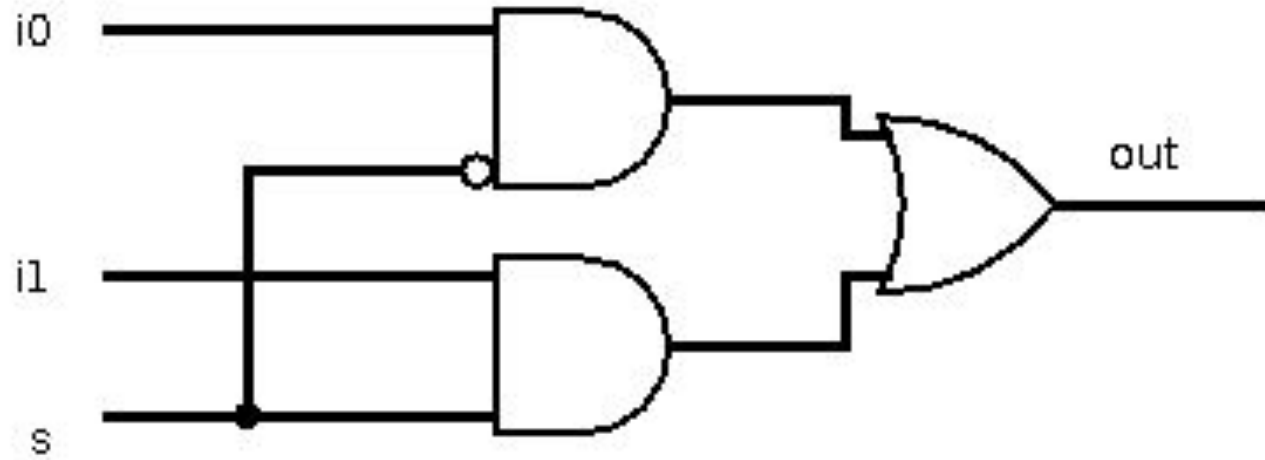
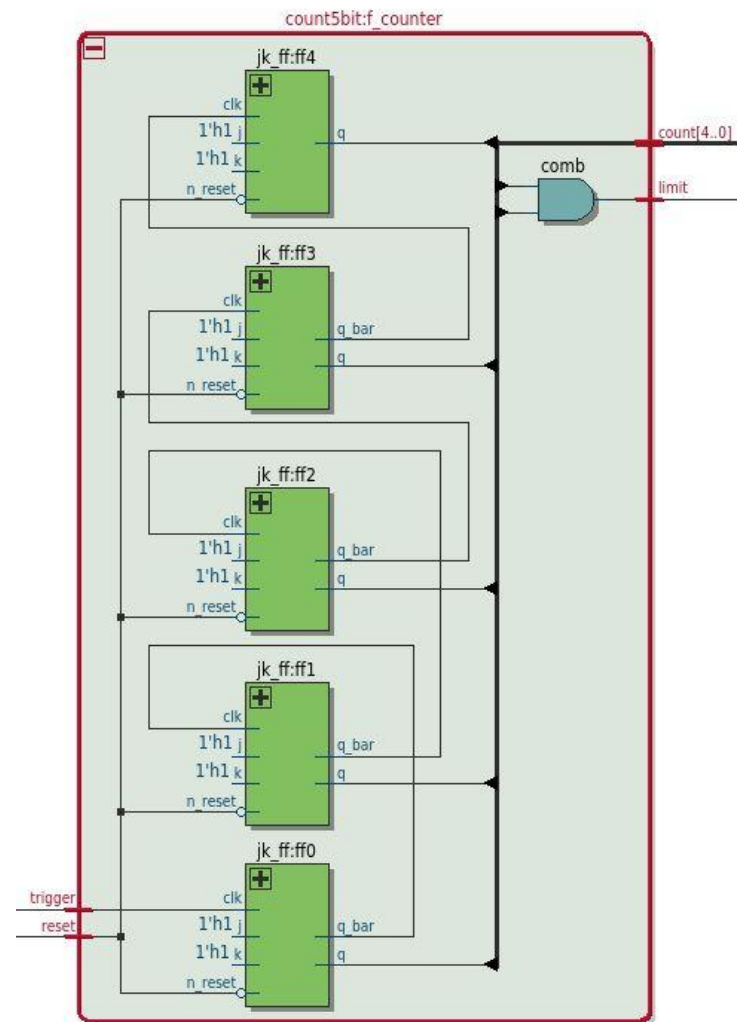


Diagrama do circuito do multiplexador de duas entradas e uma saída



RTL view do component count5bit